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Dr Johannes Wachs

Corvinus University of Budapest, Hungary

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Re: Research Assistant / PhD Student positions, Geography of Code group

Dear Dr Wachs,

A platform started my software career in 2018, and AI agents are reshaping it in 2026. I was a computer science student at FAST-NUCES in Pakistan when I took my first web-scraping jobs on Fiverr. By February 2026 I had completed more than 300 orders there and helped other students start freelance careers of their own. To pass on what I learned, I now write web-scraping-guide.com, a free public guide to web data collection. Freelancing grew into a data engineering career and remote work for employers and clients in Canada, Italy, Ireland, the United Arab Emirates, the United States and the United Kingdom. Over the past year the work itself has changed, and I now build AI into it. At M+C Saatchi Fluency I designed a self-healing agent that diagnoses and repairs our forum crawlers when a site changes, and a builder agent that creates new crawlers, while I decide what goes into production; in my own work, AI coding assistants help me write and test code. How the digital economy changes software work, where it is done and by whom, is what your group studies. I have lived that change for eight years, and I would like to spend the next years measuring it.

Your work. The paper I keep returning to is *Who is using AI to code?* with Daniotti, Feng and Neffke (*Science*, 2026). It estimates that AI wrote about 29% of US developers' Python functions by the end of 2024 and raised their output by about 3.6%, and that the gains went to experienced developers, not to the early-career developers who used AI more. That matches what I see: AI assistance helps me only because years of writing such code by hand let me judge its output quickly. The paper is open about its limits, open-source Python on GitHub in six countries, and calls for widening the analysis to countries at different income levels. Pakistan is outside that sample, and so is much of the work developers like me do, in freelance orders and private client repositories. Your paper with del Rio-Chanona and Lauretsyeva (*PNAS Nexus*, 2024) adds a caution I take seriously: Stack Overflow activity fell by about 25% after ChatGPT's release, relative to sites where it was less available, so the public traces we study are themselves changed by the technology under study. Together they point to the problem I most want to work on: measuring a change in work when that change also alters what the data can see.

Coding, modelling and data wrangling. My strongest evidence is Dubizzle Labs in Lahore (February 2022 to August 2024), the company behind the Dubizzle, Bayut, Zameen and OLX marketplaces. I joined as an Associate Software Engineer and owned one data-warehousing product end to end: its data model, its business rules and the objectives it served. I then designed the company's first web-scraping architecture on AWS, with crawlers for 15 markets collecting listings and online discussion in English, Arabic, Urdu and Hindi. On that data I built the data models and machine learning behind competitor-intelligence dashboards: what people were discussing, how much volume each site carried, and what worked on which platform and in which region. Comparing online behaviour across markets, languages and sites taught me early that the same count can mean different things on different platforms. I was promoted twice in about two and a half years, to Senior Data Engineer, led a team of three engineers in scraping and data warehousing, and helped junior colleagues into freelance and remote work; two of them got remote jobs. Since May 2023 I have also worked on contract for Fix.com, a Canadian replacement-parts business with sites across the USA, Canada and Europe, where I built OCR-based models that parse parts catalogues from PDFs and models that backfill missing parts data, and where since June 2026 I have designed and run its crawler platform alone (about 10 million rows across 82 datasets). Python and SQL are my working languages, with PySpark, PostgreSQL, Redshift and Athena when the data are large.

Measuring opinion online. I won a Chevening Scholarship for 2024/25 on that community work and on one idea: a centralised, cross-platform social media analytics system, so that government can decide on evidence of public opinion. My MSc in Data Analytics at the University of Portsmouth (Distinction) gave me deeper training in machine learning and AI, and my dissertation, supervised by Dr Ella Haig, tested the idea on about 279,000 posts about UK economic policy from X, Reddit, YouTube and Quora. Its weakness taught me most: the classifier was validated against labels seeded by a lexicon, so the check was partly circular. That became a paper with Dr Haig, *Is the Platform Part of the Measurement?*, which treats each platform as a survey mode. In a simulation I built, pooling platform scores at face value gives the wrong sign of daily aggregate sentiment on roughly a quarter of

days. We are posting it to arXiv and plan an ICWSM submission. At M+C Saatchi Fluency in London, where I have been a part-time Data Engineer since December 2024, I learned the social media industry from the inside: collecting, labelling, tagging and analysing text and sentiment for global brands in consumer goods, automotive, sport, e-commerce and health, and for UK government. I designed and operate its fleet of 64 forum crawlers, with the self-repairing agent secured by prompt fencing and approval gates, and a weekly pipeline of 2.2 million rows.

Building AI, and checking it. Since 2025 I have been co-founder and CTO of Artemis AI, a social-media intelligence start-up for creators and businesses, where I lead a team of five across models, machine learning, deployment, app development and design, and take part in pitching. The architecture we agreed on 6 October, on AWS, combines collectors on the Meta Graph API, an Apache Iceberg warehouse, a three-tier labelling cascade (our own NLP models, then DeepSeek, then Claude when confidence is low), image and video understanding, and 48 models in all. The core collection and labelling runs weekly today; an advisor on Amazon Bedrock, crisis alerts and 15 further models are being built for our beta on 15 December 2026. I went hands-on with the models because accuracy is the hard problem, and read widely to do it, from Grimmer, Roberts and Stewart's *Text as Data* to work on popularity prediction and on validating LLM annotation. That reading shapes how we build: we are designing evaluation around human-labelled data, so that every model is checked against people rather than against its own confidence or another model before it reaches users.

A question I could contribute to. How do AI coding agents change the division of labour in software work, and for whom? Agents now leave signatures in public traces: co-author trailers in commits, bot accounts that open pull requests, and the code patterns your classifier detects. Following developers' commits, collaboration networks and, through LinkedIn, their careers before and after they start working with agents, one could ask whether agent use changes who leads a project and the uneven workloads that your paper with Betti, Gallo and Battiston (*Nature Communications*, 2025) links to success; whether review work shifts towards senior developers; and whether developers in lower-income countries, including the many who enter global software work through freelance platforms, see the gains your *Science* paper finds. I would build in the habit my own research gave me: asking how much of a measured change comes from a platform's visibility rules rather than from behaviour. A trailer that is on by default for one tool and absent for another, a blank profile location, or work moving into private repositories can each look like a change in adoption. I would check trace-based measures against repositories where true agent use is known, starting with my own.

Why a PhD, and why now. I want to turn what I have learned, in the MSc at Portsmouth, in social media work at Fluency and in building Artemis AI, into research that is rigorous enough to trust. Industry taught me to build systems that collect and label online data at scale; it did not teach me to show causally what those data say about people and their work. A PhD in your group would give me that training, and I would carry the results back into how social media and data products are built and evaluated, so that my academic and professional work in this field strengthen each other.

What I do not yet bring. I have no peer-reviewed publication yet. My statistics are applied: hypothesis tests, the bootstrap, forecasting and simulation. I have not been trained in econometrics and causal inference, such as the difference-in-differences design of your Stack Overflow paper, and I have used network tools in practice without formal training in network science. I would close these gaps deliberately, through the doctoral school's courses and by applying the methods on the group's projects from my first month as a research assistant, and I would aim to take a first project with you through to a submitted paper.

Practicalities. I am available full time from 1 December 2026 and would relocate to Budapest. My referees are Dr Ella Haig, Associate Professor in AI, School of Computing, University of Portsmouth, my dissertation supervisor and co-author (Ella.Haig@port.ac.uk), and Dr Fahad Ahmad, Senior Lecturer, Centre for Cybercrime and Economic Crime, University of Portsmouth, who taught and assessed me on the MSc (Fahad.Ahmad@port.ac.uk). Thank you for considering my application.

Yours sincerely,

Asad Ikram